

What is an Image



8 → 255

What is an Image?

An image is a grid (matrix) of numbers that represents colors.

1. Image = Pixels

- An image is made of tiny dots called **pixels**

2. Each pixel has numbers

Grayscale image

- Each pixel = **one number**
- Range: 0 → 255
- 0 = black
- 255 = white

👉 Shape:

(height, width)

Color image (RGB)

- Each pixel = **3 numbers**
- Red
- Green
- Blue

👉 Example pixel:

[255, 0, 0] → Red

[0, 255, 0] → Green

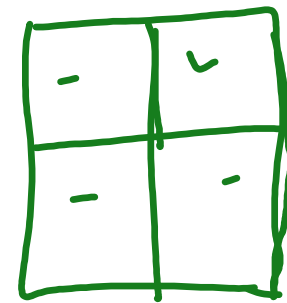
[0, 0, 255] → Blue

👉 Shape:

(height, width, 3)

Example

```
print(image.shape)
```



Channel : (3) RGB

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Output:

(224, 224, 3)

👉 Meaning:

- 224 $\rightarrow$ height
- 224 $\rightarrow$ width
- 3 $\rightarrow$ RGB channels

 One-line definition (best for class)

“An image is a matrix of pixel values, where each pixel stores color information.”

